

Assignment Two

CL 469 - Lattice Boltzmann method for fluid flow

Date: 31-03-25

Submission due date: 15-04-25

Maximum Marks: 10

Weightage: 10%

- The assignment needs to be submitted individually. You are encouraged to write your own lattice Boltzmann code. It is perfectly fine to use external sources (including snippets given in the course tutorial, ChatGPT, Deepseek, Gemini Code Assist and other online resources) to develop your version of the code. In all cases, **it is necessary** to mention which resources you have consulted in the assignment report. Failure to report the use of external resources will be treated as academic malpractice.
- A report (pdf format) and the code used for the assignment need to be submitted as a single compressed/zip file. Name the zip file as *assignment_one_your_roll_number.zip*. To keep the size of the zip file manageable, please ensure that data generated during simulations is **not** included in the zip file. The code file(s) is expected to be $\sim 10 - 100$ KB, while the report is less than ~ 5 MB. Ensure that the code compiles and produces results that are consistent with the answers/data in the pdf report.
- It is okay to discuss the questions asked in the report preparation with class fellows, the teaching assistant and the course instructor. Copying the code and answers from class fellows will be considered academic malpractice.

Problem statement: Consider a 2D plane Poiseuille flow of water under gravity through a long channel of length $L_p = 0.1$ m and width $H_p = 5 \times 10^{-5}$ m. Subscript _p denotes quantities in physical (SI) units. For simplicity, we investigate the system when it has reached a steady state. Gravity is acting in the x -direction, and is switched on at $t = 0$. The gravitational acceleration is $\mathbf{g}_p = (g_{x,p}, g_{y,p}) = (9.8, 0)$ m/s². Kinematic viscosity of water is $\nu_p = 10^{-6}$ m²/s and density of water is $\rho = 10^3$ Kg/m³. We can easily obtain the velocity profile and force per unit length of the channel considering a small section of the channel for the analysis. Fig. (1), shows one such section of the channel. The length of the section of the channel is $L_s = H$. The top and bottom boundaries of the channel are walls where the no-slip boundary condition is applied. To model the behaviour of the fluid along such a long channel, we apply periodic boundary conditions at the left and right boundaries of the channel section.

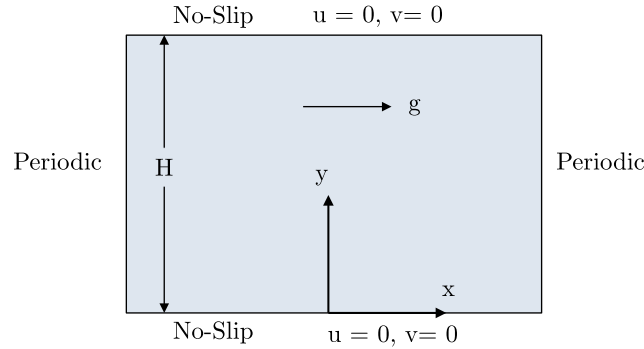


Figure 1:

The mass and momentum balance equations are

$$\frac{\partial \rho_p}{\partial t} + \nabla \cdot (\rho_p \mathbf{u}) = 0 \quad (1)$$

$$\rho_p \left(\frac{\partial \mathbf{u}_p}{\partial t} + \mathbf{u}_p \cdot \nabla \mathbf{u}_p \right) = -\nabla p_p + \mu_p \nabla^2 \mathbf{u}_p + \rho_p \mathbf{g}_p \quad (2)$$

with the equation of state

$$p_p = \rho_p c_s^2 = \rho_p R_p T_p, \quad (3)$$

where p_p is pressure, $\mathbf{u}_p = (u_{x,p}, u_{y,p})$ is the velocity field, ρ_p denotes fluid density, ∇_p denotes spatial gradient, R_p is specific gas constant and T_p is temperature.

As mentioned before, in most part of the channel no appreciable changes in velocity take place along the length of the channel. In this state, also known as the fully developed regime, all gradients of velocity in x -direction are zero. Further, no-slip boundary conditions at static top walls and bottom walls render $u_{y,p} = 0$. Using these conditions in Eq. (2) and further assuming steady state condition, the momentum balance becomes

$$\mu_p \frac{\partial^2 u_{x,p}}{\partial y^2} + \rho_p g_{x,p} = 0, \quad (4)$$

where μ_p is the dynamic viscosity.

Integrating Eq. (4) w.r.t to y , and applying boundary conditions $u_{x,p}(y = 0) = 0$ and $u_{x,p}(y = H_p) = 0$, we get the u_x velocity profile as

$$u_{x,p} = \frac{g_{x,p}}{2\nu_p} y(H_p - y), \quad (5)$$

where $\nu_p = \frac{\mu_p}{\rho_p}$. We can define Reynolds number as $Re = \frac{U_p H_p}{\nu_p}$, where U_p is the characteristic velocity scale. One can define multiple characteristic velocity scales from Eq. (5) such as the maximum velocity in the channel or height averaged velocity. The maximum velocity in the channel can be shown to occur at $y = H_p/2$ and is given as $u_{x,p}^m = \frac{g_{x,p} H_p^2}{8\nu_p}$. Similarly the height averaged velocity can be written as

$$\bar{u}_{x,p} = \frac{1}{H_p} \int_0^{H_p} u_{x,p}(y) dy.$$

We choose the maximum velocity as the characteristic velocity scale, since it also helps us easily identify the maximum Mach number we encounter in our simulations. Thus, the Reynolds number in the given problem is

$$Re = \frac{u_{x,p}^m H_p}{\nu_p} = 0.1531.$$

Now, we can also obtain the wall shear stress acting on the bottom wall as

$$\tau_w = \mu \left. \frac{\partial u_{x,p}}{\partial y} \right|_{y=0} = \frac{\rho_p g_{x,p} H_p}{2}. \quad (6)$$

Thus, the force acting on bottom plate is given as

$$\begin{aligned} F_p &= \int_A \tau_w dA \\ &= \int_0^{\Delta z_p} \int_0^{L_s} \tau_w dx dz \\ &= \frac{\rho_p g_{x,p} H_p L_s \Delta z_p}{2}, \end{aligned} \quad (7)$$

which acts in the positive x -direction. This makes sense because as the fluid moves in the channel, it tries to drag the plate with it thus imparting force in positive x -direction. Similarly, one can also find the same force acting on the top plate as well.

Note, the force acting on bottom wall as given in Eq. (7) is half of the weight of the fluid in the channel. We also see viscosity does not enter the expression for force. This has a rather surprising consequence which can be seen by non-dimensionalizing the force using the weight of the fluid in the channel. The non-dimensional force is thus given as

$$\tilde{F} = \frac{F_p}{\rho_p g_{x,p} H_p L_s \Delta z_p} = 0.5. \quad (8)$$

We see the non-dimensional force has no dependence on Reynolds number, which is surprising because this suggests that if Reynolds number is set to zero, we should still have some non-zero force on the wall. Also, if viscosity is set to zero, the non-dimensional force should still remain the same according to Eq. (8).

To explain the viscosity independence of the force experienced by the wall, note that the viscous stress is proportional to the viscosity and velocity gradient while the velocity itself is inversely proportional to the viscosity, Eq. (6).

In this assignment you will simulate this system and measure the forces acting on a plate through various approaches.

Choice of simulation parameters and corresponding conversion factors (sections 7.1 and 7.2 textbook):

General considerations

1. The time step $\Delta t = 1$ and grid spacing $\Delta x = 1$ are fixed. Simulation parameters are always reported in lattice units, which are dimensionless. For example, relaxation parameter $\tau = 1$, kinematic viscosity $\nu = 1$, density $\rho = 1$, gravity component $g = 10^{-6}$ and so on.
2. For the moment, we know only the simple BGK collision operator. For simple flows, BGK operator works for relaxation parameters in the range $0.55 < \tau < 1.5$. The lower limit on τ is related to the stability, while the upper limit results from slip on solid walls. It may, of course, work outside the said range, too, if the flow is unidirectional.
3. The problems we are interested in this course are incompressible/weakly compressible flow situations. Thus, we do not match the Mach number in the real world and in our simulations. The current problem involves only steady state with $\mathbf{u} = 0$ resulting in zero Reynolds Number $Re = 0$ and Mach number $Ma = 0$. The Mach number in simulations is lattice Mach number $Ma = \frac{|\mathbf{u}|}{c_s}$ has to be kept much smaller than 1 at all times. This means that although the Mach number is zero at steady-state, we have to ensure that at no point (in space and time) the fluid velocity becomes comparable to the lattice speed of sound. A general rule of thumb is to keep Ma less than 10^{-2} or fluid velocity magnitude $|\mathbf{u}|$ less than 10^{-3} at all times.
4. Unlike in the previous assignment, in this problem the maximum possible velocity that can occur in the system can be defined analytically. In lattice units it is given as $u^m = \frac{gH^2}{8\nu}$. As we define Re in terms of maximum velocity, one can choose a τ based on the stability criteria and a grid resolution H , and find the maximum velocity in the channel. Provided the maximum velocity (and hence Ma) is well below the maximum permissible value, the corresponding value of gravity can be found.

For the current problem

Let C_l, C_t and C_ρ be the conversion factors. Physical units are denoted with a subscript _p and all the quantities in lattice units are denoted without any subscript.

1. BGK operator allows a small window of possible kinematic viscosity values. Given that we are working with the BGK operator, the kinematic viscosity then sets the constraint for the choice of simulation parameters. Choose the kinematic viscosity $\nu = 0.1$ that corresponds to relaxation parameter $\tau = \frac{1}{2}\nu + 0.5\Delta t = 0.8$.

$$\nu \frac{C_l^2}{C_t} = \nu_p = 10^{-6} \text{ m}^2/\text{s} \quad (9)$$

2. We choose the width of the channel in y -direction to be $H = 11$. This means we have chosen 10 grid points to represent a parabola.

$$HC_l = H_p = 5 \times 10^{-5} \text{ m}, \quad (10)$$

3. The dimensionless number we have to match is the Reynolds number

$$\begin{aligned} \text{Re} &= \frac{u^m H}{\nu} = 0.1531 \\ \implies u^m &= 1.392 \times 10^{-3}. \end{aligned} \quad (11)$$

The lattice speed of sound is fixed with the D2Q9 lattice to $c_s = 1/\sqrt{3}$. Thus, the Mach number turns out to be

$$\text{Ma} = \frac{u^m}{c_s} = 2.41 \times 10^{-3}$$

which is sufficiently small for the simulations

4. Finally we can obtain gravity from the analytical expression for maximum velocity as

$$\begin{aligned} u^m &= \frac{gH^2}{8\nu} \\ \implies g &= 9.203 \times 10^{-6}. \end{aligned} \quad (12)$$

5. In this case, the viscous time scale decides the time taken by the system to reach steady state. The viscous time scale is given as

$$t_\nu = \frac{H^2}{\nu} = 1210. \quad (13)$$

General rule of thumb is to simulate the system atleast upto 10 time the viscous time scale. Thus, the total simulation time should be $> 10t_\nu = 12100$.

6. Density (mass) conversion

$$\rho C_\rho = \rho_p, \quad (14)$$

where $\rho = 1$ in lattice units.

Note that only the Reynolds number was matched in our simulations with the one given in the physical system. For the incompressible flow that we have targeted, the Mach number is negligible (≈ 0). In our simulations though, the Mach number is small but not negligible. Question is, does this mean that the results we obtain such as forces do not correspond to the physical situations? No, as we can see the force does not depend on Mach number. In this special case it is also independent of Reynolds number. In a general case where the force on solid bodies are dependent on Reynolds number, Mach number still doesn't play a role in determination of the forces. Thus, as long as Mach number is small enough to guarantee stability of the simulations, the forces obtained are independent of Mach number.

Now, the last bit is to understand how to calculate forces on solid bodies.

Forces on plate:

As discussed in class, forces on a solid body due to fluid can be calculated by projecting the stress tensor $\boldsymbol{\sigma}$ on the solid surface and integrating over the entire surface area as

$$\mathbf{F}_s = \int_A (\boldsymbol{\sigma} \cdot \mathbf{n}_s) dA, \quad (15)$$

where $\mathbf{n}_s$ is the outward normal to the solid body. The stress tensor can be further divided into two parts as

$$\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\tau}, \quad (16)$$

where $p = \rho c_s^2$ is the pressure, $\mathbf{I}$ is the identity tensor and $\boldsymbol{\tau}$ is the viscous stress tensor. In lattice Boltzmann simulations, the viscous stress tensor can be directly calculated from the non-equilibrium part of the populations since they already contain information about fluid velocity gradients. Thus, the viscous stress tensor can be computed as

$$\tau = \tau_{\alpha\beta} = \left(1 - \frac{\Delta t}{\tau}\right) \sum_i c_{i\alpha} c_{i\beta} (f_i - f_i^{\text{eq}}), \quad (17)$$

where c_i denotes the lattice velocity in i^{th} lattice direction. In this simple case, as the flow is unidirectional and fully developed, the viscous stress tensor has only one unique component which is

$$\tau_{xy} = \mu \frac{\partial u_x}{\partial y}. \quad (18)$$

However, we have already discussed that integrating the stress tensor over the solid surface can be cumbersome for arbitrarily shaped solids. Hence, we can simplify the calculation of forces by using the Momentum Exchange Algorithm (**ref. textbook 5.4.3**). At fluid node near a solid boundary, the change in momentum due to the presence of a wall can be written as

$$\Delta \mathbf{p} = \sum_{x_{i,w}} \mathbf{c}_i f_i^{\text{in}} - \mathbf{c}_{\bar{i}} f_{\bar{i}}^{\text{out}}, \quad (19)$$

where i denotes the lattice direction towards the solid wall such that $\mathbf{c}_{\bar{i}} = -\mathbf{c}_i$, f_i^{in} denotes the population streaming from fluid to wall and $f_{\bar{i}}^{\text{out}}$ denotes the population streaming from solid to wall. In the notation introduced in the course, f_i^{in} is the post-collision population also denoted as f_i^* , and $f_{\bar{i}}^{\text{out}}$ is the bounced back population $f_{\bar{i}}$. Thus, Eq. (19) can then be written as

$$\Delta \mathbf{p} = \sum_{x_{i,w}} \mathbf{c}_i f_i^* + \mathbf{c}_{\bar{i}} f_{\bar{i}}, \quad (20)$$

where halfway bounce back states $f_{\bar{i}} = f_i^* - 2w_i \rho_w \frac{\mathbf{c}_i \cdot \mathbf{u}_w}{c_s^2}$, with ρ_w being the fluid density near wall and $\mathbf{u}_w$ being the wall velocity. Since, the walls are static in the current problem, $\mathbf{u}_w = \mathbf{0}$. Thus, for this very simple case, halfway bounce back states $f_{\bar{i}} = f_i^*$. This simplifies Eq. (20) to

$$\Delta \mathbf{p} = \sum_{x_{i,w}} 2\mathbf{c}_i f_i^*. \quad (21)$$

Thus force on the wall can be computed by summing up all the differential changes in momentum per unit time over all the fluid nodes that are neighbour to the solid wall (denoted by $\mathbf{x}_b$) as

$$\mathbf{F}_s = \frac{\Delta \mathbf{p} (\Delta x)^2}{\Delta t} = \frac{(\Delta x)^2}{\Delta t} \sum_{\mathbf{x}_b} \sum_{x_{i,w}} (\mathbf{c}_i f_i^* + \mathbf{c}_{\bar{i}} f_{\bar{i}}) = \frac{(\Delta x)^2}{\Delta t} \sum_{\mathbf{x}_b} \sum_{x_{i,w}} 2\mathbf{c}_i f_i^*. \quad (22)$$

lattice Boltzmann algorithm:

1. Initial fluid velocity $\mathbf{u}(\mathbf{x}, t = 0)$ and density $\rho(\mathbf{x}, t = 0)$ distribution are given everywhere. Equilibrium fluid populations at time $t=0$ can then be found. Initialize the fluid populations with equilibrium fluid populations. In this problem, the initial velocity is zero everywhere and density is uniform ρ_0

$$f_i(\mathbf{x}, t = 0) = f_i^{\text{eq}}(\mathbf{x}, t = 0) = w_i \rho_{lb} \quad (23)$$

2. Time iteration loop

(a) Update density and velocity

$$\rho = \sum_i f_i \quad (24)$$

$$\rho \mathbf{u} = \sum_i f_i \mathbf{c}_i + \frac{\Delta t}{2} \mathbf{F} \quad (25)$$

Note the presence of a force term (force per unit volume) on the right-hand side of Eq. (25). We will discuss the inclusion of force terms in the Chapman-Enskog analysis in the coming sessions. How to code force inclusion is discussed in the class. In the current problem $\mathbf{F} = \rho \mathbf{g}$ and $\mathbf{F} \neq \mathbf{g}$.

- (b) Compute the viscous stress tensor using Eq. (17) which will be required to calculate forces. After the total stress tensor is calculated, compute forces separately on top and bottom wall. Alternatively, at this step one can also apply momentum exchange algorithm given in Eq. (22) to obtain the forces acting on the top and bottom plates.
- (c) Collide fluid populations

$$f_i^* = \left(1 - \frac{\Delta t}{\tau}\right) f_i + \frac{\Delta t}{\tau} f_i^{\text{eq}} + F_i \quad (26)$$

Here, we have used the BGK collision operator, and f_i^* are post-collision fluid populations. We use the Guo scheme to include the force term

$$F_i = \left(1 - \frac{\Delta t}{2\tau}\right) w_i \left(\frac{\mathbf{c}_i - \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})}{c_s^4} \mathbf{c}_i \right) \cdot \mathbf{F} \Delta t, \quad (27)$$

$\mathbf{F}$ in the Eq. (27) and Eq. (25) are the same. Similarly $\mathbf{F}$ in the Eq. (27) and in the lattice force terms F_i in Eq. (27) and Eq. (26) are the same.

- (d) Propagate fluid populations

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i^*(\mathbf{x}, t) \quad (28)$$

- (e) Apply boundary conditions
we have discussed the no-slip (bounce-back) and periodic boundary conditions. Sufficient for this problem. Periodic boundaries at $x = 0$ and $x = L_x$ and bounce-back at top ($y = L_y$) and bottom ($y = 0$).

3. Exit the code

Report preparation:

1. For the reference parameters given here, compare the steady state velocity profile in simulations with the analytical solution given in Eq. (5). Note that, the plot should be in terms of non-dimensional variables. The reference parameters are - $\tau = 0.8$, $\nu = 0.1$, $H = 11$ and $g = 9.203 \times 10^{-5}$. Also check what is the maximum value of y component of velocity in the simulations.
2. Obtain the forces acting on the top and bottom wall using Momentum Exchange method (Eq. (22)). Non-dimensionalize the forces in x -direction obtained from simulations and compare with Eq. (8). Note that both the force components in x and y direction are non-zero. What is the force in y direction? Compare it with simulations.
3. Remember that we said that the forces in incompressible flows like the current one are not affected by Mach number. Hence, even though the Mach number in our simulations are much higher than the Mach number in the physical system, the forces should not change. To verify this, run three simulations with three different Mach numbers (must be smaller than 10^{-2}), and compute the forces on top and bottom wall. Show that the non-dimensional forces do not change with changing Mach numbers. To change Mach number, the maximum velocity must be changed in the simulations. This can be done by altering the grid resolution. But, ensure the Reynolds number is the same. **Also, since in this case the shear force on the wall is independent of Reynolds number, the non-dimensional forces should still stay the same for different Reynolds number in simulations.**
4. Calculate the viscous stress tensor from Eq. (17) everywhere in the fluid. In 2D, the viscous stress tensor has four components

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_{xx} & \tau_{xy} \\ \tau_{yx} & \tau_{yy} \end{bmatrix} = \mu \begin{bmatrix} 2 \frac{\partial u_x}{\partial x} & \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) \\ \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) & 2 \frac{\partial u_y}{\partial y} \end{bmatrix} = \quad (29)$$

For the flow we have considered, τ_{xx} and τ_{yy} must be negligible. Furthermore, since the viscous stress tensor is symmetric, $\tau_{xy} = \tau_{yx}$. Verify that this is really the case in the simulations.

5. With the viscous stress tensor obtained from simulations, construct the total stress tensor $\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\tau}$. Obtain the force on the top and bottom wall separately by projecting the stress tensor on to the walls and integrating. For example, consider the bottom wall. The outward normal to the solid wall is $\mathbf{n}_{s,b} = (0, 1)$. Thus, the force on the bottom wall can be obtained numerically as

$$\mathbf{F}_{s,b} = \sum_{x=0}^{L_s} \boldsymbol{\sigma} \cdot \mathbf{n}_{s,b} \Delta x \Delta z. \quad (30)$$

To obtain the forces take $\Delta x = \Delta z = 1$.

Also, for this simple case one can obtain force on a wall simply by computing

$$\mathbf{F}_{s,b} = \sum_{x=0}^{L_s} \mu \left(\frac{du_x}{dy} \right)_{y=0}. \quad (31)$$

To this end, also obtain the force on top and bottom walls using Eq. (31) at steady state. Since, the velocity field is known at steady state, use forward/backward difference to obtain $\frac{du_x}{dy}$.

Compare the forces obtained from Momentum Exchange (Eq. (22)), integrating the stress tensor (Eq. (30)) and by finite difference method (Eq. (31)) for various channel resolutions $H = 5, 10, 20$ and 40. Plot two graphs for each wall, where the x -axis represents the channel width in lattice units and y -axis represents the relative % error in non-dimensional forces from the analytical value (Eq. (8)). Note, relative % error is defined as

$$\epsilon = \frac{F_{s,x} - 0.5}{0.5} \times 100 \text{ \%}.$$

6. As we have discussed in the class, the halfway bounce back coupled with BGK collision operator provides a narrow operating range of relaxation parameter. While the lower limit of $\tau > 0.5$ is related to stability, the upper limit is related to slip velocity at the wall. Recall, that we have discussed that the location of the solid wall is between the last fluid node and the first solid node. We have also found that the velocity at the bottom wall for BGK collision operator coupled with halfway bounce back for a channel flow can be expressed as

$$u_x(y_w) = u_x(y_b) - \frac{\Delta x}{2} \frac{\partial u_x}{\partial y} \Big|_{y=y_b} + \frac{2c^2}{3} \left(\tau - \frac{\Delta t}{2} \right)^2 \frac{\partial^2 u_x}{\partial y^2} \Big|_{y=y_b} + \mathcal{O}(\Delta x^3). \quad (32)$$

For the wall location to be exactly $\Delta x/2$ distance away from the last fluid node, $\tau = \Delta t \left(1 + \sqrt{\frac{3}{16}} \right)$.

Thus, BGK operator gives viscosity dependent wall location in the simulations. At high viscosity values this results in a non-zero slip velocity at the wall which is unphysical.

To show this in simulations, simulate the same channel flow problem with all the reference case parameters except τ . Increase τ to 2 and 5 and show that the velocity at the solid wall is non-zero.

This artefact can be resolved by using advanced collision operators like the Two Relaxation Time (TRT) collision operator, which through proper tuning can give viscosity independent wall location. To this end implement the TRT collision operator (**ref. to Book section 10.6.2**) given as

$$f_i^* = f_i - \frac{\Delta t}{\tau^+} (f_i^+ - f_i^{\text{eq}+}) - \frac{\Delta t}{\tau^-} (f_i^- - f_i^{\text{eq}-}), \quad (33)$$

where

$$\begin{aligned} f_i^+ &= \frac{f_i + \bar{f}_i}{2}, & f_i^- &= \frac{f_i - \bar{f}_i}{2}, \\ f_i^{\text{eq}+} &= \frac{f_i^{\text{eq}} + \bar{f}_i^{\text{eq}}}{2}, & f_i^{\text{eq}-} &= \frac{f_i^{\text{eq}} - \bar{f}_i^{\text{eq}}}{2} \end{aligned}$$

where $\tau^+ = \frac{\nu}{c_s^2} \frac{\Delta t}{2} + 0.5$ and τ^- is a free parameter. Use $\tau^+ = 2$ and 5 and show that using TRT collision operator coupled with halfway bounce back one can get rid of the unphysical slip at the boundary.

7. At steady state one can show that the work done by gravity is equal to the viscous dissipation. The viscous dissipation in 2D is given as

$$W_d = \mu \int_V \left[2 \left\{ \left(\frac{\partial u_x}{\partial x} \right)^2 + \left(\frac{\partial u_y}{\partial y} \right)^2 \right\} + \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} \right)^2 \right] dV, \quad (34)$$

where V denotes a volume integral. For the current system, Eq. (34) can be simplified to write

$$W_d = \mu \int_V \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} \right)^2 dV. \quad (35)$$

Now, recall from Chapman-Enskog analysis,

$$\Pi_{xy} = -\rho c_s^2 \tau \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} \right), \quad (36)$$

and that $\mu = \rho \left(\tau - \frac{\Delta t}{2} \right) c_s^2$. From this, we can write viscous dissipation as

$$W_d = \frac{\tau - \Delta t/2}{\rho c_s^2 \tau^2} \int_V \Pi_{xy}^2 dV. \quad (37)$$

where $\Pi_{xy} = \sum_i (f_i - f_i^{\text{eq}}) c_{ix} c_{iy}$.

Analytically, one can write the viscous dissipation as

$$W_{d,p} = \mu \int_V \left(\frac{\partial u}{\partial y} \right)^2 dV, \quad (38)$$

where putting $u_x = \frac{g}{2\nu} y(H - y)$, we get

$$W_{d,p} = \frac{1}{12} \rho_p^2 g_{x,p}^2 L_s H_p^3 \Delta z_p. \quad (39)$$

Through simulations obtain viscous dissipation using Eq. (37) and compare it with the analytical value obtained from Eq. (39). It is easier to compare the non-dimensional dissipation which can be found through appropriate scaling. Furthermore, compare the simulations with analytical value for varying channel widths $H = 5, 10, 20$ and 40. Does the accuracy improve with increasing resolution?

Also, show that the work done by gravity is analytically equal to the viscous dissipation given in Eq. (39).